Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C++, JavaScript, SQL

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Concurrency & Multithreading

Backend & Software Development: RESTful APIs, FastAPI, Flask, Node.js, API Design, Socket Programming, JWT/OAuth, API Testing

Databases & Storage: MySQL, SQLite, Redis, SQL Query Optimisation, Caching

Software Engineering: System Design, Distributed Systems, Unit Testing, PyTest, Git, GitHub

AI/ML & Data: Machine Learning, RAG, NLP, Generative AI, Scikit-learn, Pandas, NumPy, Data Analysis, Power BI

EXPERIENCE

VOIS for Tech University Engagement Program (Virtual Internship) **|LINK** — ***Aug – Sep, 2025***

- Engineered a modular data processing pipeline to preprocess, analyse, and query two real-world datasets containing 8,500+ records using LLM-powered natural language interfaces.
- Designed and implemented automated validation and response extraction workflows, improving the consistency and reliability of generated outputs across diverse user queries.
- Optimised data processing by integrating preprocessing, prompt generation, response parsing, and result validation into a reusable workflow, reducing manual analysis effort and improving scalability.

PROJECTS

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, persistence, and crash recovery, validating the correctness and reliability of the storage and networking components.

API Rate Limiter Library **|LINK**

- Engineered a Python API rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API traffic control.
- Designed a modular architecture supporting 2 storage backends, InMemory and Redis, with thread-safe concurrent request handling and reusable abstractions for flexible deployment.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack expense management platform using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP-based parsing to automate extraction of 10+ receipt fields, including merchant, date, amount, and category.
- Engineered 15+ REST API endpoints with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export, supporting secure end-to-end expense management workflows.
- Deployed the application across Render and Vercel, integrating 2 authentication mechanisms, frontend-backend communication, database persistence, and production configuration for a complete full-stack application.

Docwise **|LINK**

- Developed Docwise, an AI-powered PDF question-answering system using Google Gemini and a lightweight RAG pipeline, processing 200+ page documents through automated text extraction, chunking, retrieval, and generation.
- Implemented a retrieval pipeline using 500-word chunks, TF-IDF vectorisation, cosine similarity, prompt engineering, and session-based chat history to retrieve relevant document context before generating answers.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — **2026**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate| **LINK** — **2026**
- Accenture Software Engineering Job Simulation | **LINK** — **2025**